1. Clinical Question & Core Hypothesis

Clinical Question

Ontologist synthesis asks why CRPC tumors show persistent AR pathway activity and therapy resistance despite AR-targeted therapy and whether transcriptional/epigenetic reprogramming (full-length AR mRNA mutations, MGMT, KRT7P1 activation) underlies this. 2) Tier 1 paths used: (a) full-length AR mRNA alterations linked to enzalutamide resistance — chosen because direct AR-transcript alterations are a primary, mechanistic route to antiandrogen failure; (b) KRT7P1 epigenetic activation under PARP/epigenetic perturbation — chosen because epithelial transcriptional reprogramming can reflect cell-state shifts that sustain malignant programs. I built the core claim on these two because they provide a mechanistic route (transcript-level AR robustness + epigenetic epithelial state change) that can be modulated by the tumor microenvironment. 3) Tier 2 evidence supporting the claim: STAT3/SOX2–driven transcriptional plasticity (supporting stemness and transcriptome remodeling) and transcriptome assembly/transfrag analyses indicate that therapy (Olaparib/cabazitaxel) can induce novel/altered transcripts and regulatory programs; these reinforce that microenvironmental cues could preserve aberrant full-length AR transcripts and activate keratin-related programs. 4) Endpoint choice: I selected therapy_response (enzalutamide response) because the strongest clinical problem highlighted by the Tier 1 paths is failure of AR-directed therapy driven by AR transcript alterations; therapy response is the most direct, actionable endpoint for guiding treatment selection. 5) Deliberate exclusions: I excluded MGMT mRNA from the core clause to keep the mechanistic claim sharp (MGMT is more relevant to PARP/DDR context and would dilute focus on AR-driven enzalutamide resistance). I also excluded broader transcriptional assembly concepts (transfrag/Oases) from the core hypothesis and placed them in secondary_modifiers as supportive methodological/contextual evidence?

Working Hypothesis

High stromal-to-tumor ratio (TSR) in castration-resistant prostate cancer (CRPC) creates a paracrine/epigenetic niche that preserves AR signaling under enzalutamide by two non–mutually exclusive mechanisms: (1) stromal-derived cues enhance stability and processing of full-length AR mRNA, and (2) stromal-epigenetic signaling activates KRT7P1, facilitating regulatory networks that sustain the AR transcriptional program. Consequently, tumors with high TSR exhibit continued AR target gene activity and reduced response to enzalutamide, defining a clinically actionable resistant subset. This model predicts: (a) TSR-high tumors will have higher levels of full-length AR mRNA relative to AR variants and elevated KRT7P1 expression; (b) stromal signals (e.g., IL-6/STAT3 axis) will increase AR mRNA stability and KRT7P1 activation; (c) disrupting stromal signaling or KRT7P1 function will restore sensitivity to enzalutamide or synergize with AR blockade. Experimental approach to test this: (i) analyze patient datasets to correlate TSR proxies with full-length AR mRNA and KRT7P1, AR target gene signatures, and clinical enzalutamide response; (ii) in vitro co-culture of prostate cancer cells with CAFs or stromal-conditioned media to assess AR mRNA stability (actinomycin D chase), processing, and AR target gene expression under enzalutamide; (iii) manipulate KRT7P1 by CRISPRi/ASO to evaluate effects on AR signaling and enzalutamide sensitivity; (iv) use organoid and PDX models with high TSR to test combination strategies targeting stromal signaling (STAT3 inhibitors), epigenetic modulation (DZNep), or PARP inhibitors (Olaparib) to overcome resistance; (v) validate imaging-phenotype correlations using TSR measurements with RNA in situ hybridization for full-length AR and KRT7P1. Outcomes supporting the hypothesis would include linkage of high TSR with preserved AR signaling despite enzalutamide and attenuation of this resistance when KRT7P1 or stromal signals are inhibited.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	full-length AR mRNA, KRT7P1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

full-length AR mRNA: full-length AR mRNA alterations linked to enzalutamide resistance — chosen because direct AR-transcript alterations are a primary, mechanistic route to antiandrogen failure; (b) KRT7P1 epigenetic activation under PARP/epigenetic perturbation — chosen because epithelial transcriptional reprogramming can reflect cell-state shifts that sustain malignant programs.

KRT7P1: KRT7P1 epigenetic activation under PARP/epigenetic perturbation — chosen because epithelial transcriptional reprogramming can reflect cell-state shifts that sustain malignant programs.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MGMT mRNA	Supporting modifier from knowledge graph
Secondary	STAT3	Supporting modifier from knowledge graph
Secondary	SOX2	Supporting modifier from knowledge graph
Secondary	Olaparib	Supporting modifier from knowledge graph
Secondary	DZNep	Supporting modifier from knowledge graph
Secondary	cabazitaxel	Supporting modifier from knowledge graph
Secondary	transfrag	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MGMT mRNA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- STAT3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.